

LECTURE NOTES MATH255

SVEN SANDFELDT

1. LECTURE 5

In this class we cover Fermat's little theorem and Euler's generalization of Fermat's little theorem.

1.1. Fermat's little theorem. Let p be a prime and consider the field $\mathbb{Z}_p$. Since $\mathbb{Z}_p$ is a field, $\mathbb{Z}_p^\times = \mathbb{Z}_p \setminus 0$ is a group under multiplication. Indeed, $1 \in \mathbb{Z}_p^\times$ serves as an identity element, and for any $a \in \mathbb{Z}_p^\times$ we know that there is some $a^{-1} \in \mathbb{Z}_p^\times$ such that $aa^{-1} = a^{-1}a = 1$, since any non-zero element in a field has a multiplicative inverse. That is, $\mathbb{Z}_p^\times$ is indeed a group with $p-1$ elements. Recall that, by Lagrange's theorem, the order of any element in a group G divides the order of the group $|G|$. As a consequence, if G is a finite group and $g \in G$ has order n then $|G| = mn$ for some integer m , and we have $g^{|G|} = (g^n)^m = e$. Applying this observation to $\mathbb{Z}_p^\times$ we conclude that if $a \in \mathbb{Z}_p^\times$ then:

$$(1.1) \quad a^{|\mathbb{Z}_p^\times|} = a^{p-1} = 1$$

where we recall that these equalities are understood in $\mathbb{Z}_p$. This is a formulation of Fermat's little theorem:

Theorem 1.1. *Let p be a prime and let $a \in \mathbb{Z}$ be any integer that is not divisible by p . Then $a^{p-1} - 1$ is divisible by p (or more commonly: $a^{p-1} \equiv 1 \pmod{p}$).*

Proof. Note that if p does not divide a then $a + p\mathbb{Z} \in \mathbb{Z}_p \setminus 0$. From the argument before the theorem it follows that: $(a + p\mathbb{Z})^{p-1} = 1 + p\mathbb{Z}$, but recalling $(a + p\mathbb{Z})^{p-1} = a^{p-1} + p\mathbb{Z}$ the theorem follows. ■

Corollary 1.1.1. *If p is prime and $a \in \mathbb{Z}$ then $a^p \equiv a \pmod{p}$.*

Proof. If a is not divisible by p then the corollary is immediate from Fermat's little theorem. If a is divisible by p then a^p and a are both equal to 0 modulo p , so the corollary also holds true. ■

As a consequence of Fermat's little theorem we can calculate congruences of certain large powers:

Example 1.1. Consider $8^{103} \in \mathbb{Z}$. We want to calculate the remainder of 8^{103} when we divide by 13. Since 13 is prime, we can use Fermat's little theorem. By Fermat's little theorem we have:

$$(1.2) \quad 8^{13-1} = 1 \pmod{13}$$

or $8^{12} = 1 \pmod{13}$. Notice that $12 \cdot 8 = 96$ so we have $8^{96} = (8^{12})^8 = 1 \pmod{13}$. It follows that $8^{103} = 8^7 \cdot 8^{96} = 8^7 \pmod{13}$. Rewriting $8 = 2^3$ we can write $8^7 = 2^{21}$ and again using Fermat's little theorem we have:

$$(1.3) \quad 8^7 = 2^{21} = 2^9 \cdot 2^{12} = 2^9 \pmod{13}.$$

Since $2^4 = 16$ we note that $2^9 = 2^4 \cdot 2^4 \cdot 2 = 16 \cdot 16 \cdot 2$ and $16 = 3 \pmod{13}$ then implies:

$$(1.4) \quad 8^{103} = 2^9 = 16 \cdot 16 \cdot 2 = 3 \cdot 3 \cdot 2 = 18 = 5 \pmod{13}.$$

Example 1.2. Show that $2^{11213} - 1$ is not divisible by 11. To see this, we note that 11 is prime, which by Fermat's little theorem implies $2^{10} = 1 \pmod{11}$. Noting that $11213 = 11210 + 3$ and 11210 is divisible by 10 we obtain:

$$(1.5) \quad 2^{11213} - 1 = 2^3 \cdot 2^{11210} - 1 = 2^3 - 1 = 8 - 1 = 7 \pmod{11}$$

and since $7 \neq 0$ in $\mathbb{Z}_{11}$ we conclude that $2^{11213} - 1$ is not divisible by 11.

Example 1.3. The integer $n^{33} - n$ is divisible by 15 for all $n \in \mathbb{Z}$. Assume first that n is not divisible by 3, applying Fermat's little theorem with the prime 3 we have $n^{33} = n \cdot (n^2)^{16} = n \pmod{3}$ so $n^{33} - n = 0 \pmod{3}$ which shows that $n^{33} - n$ is divisible by 3. If n is divisible by 3 then n and n^{33} are both divisible by 3 so $n^{33} - n$ is divisible by 3. Either way, $n^{33} - n$ is always divisible by 3. If n is divisible by 5 then n and n^{33} are divisible by 5 so $n^{33} - n$ is divisible by 5. If n is not divisible by 5 then we apply Fermat's little theorem with the prime 5 and obtain:

$$(1.6) \quad n^{33} = n \cdot (n^4)^8 = n \pmod{5}$$

so, again, we have $n^{33} - n = 0 \pmod{5}$, and $n^{33} - n$ is divisible by 5. So, we have shown that $n^{33} - n$ is divisible by 3 and 5 for all integers n , which implies that $n^{33} - n$ is divisible by $3 \cdot 5 = 15$ for all integers n .

1.2. Euler's generalization of Fermat's little theorem. Before formulating Euler's generalization of Fermat's little theorem it is worthwhile to introduce the *unit group* of a ring with unity.

Definition 1.1. Let R be a ring with unity. We define the unit group of R , denoted $R^\times$, as the set of all units in R .

To motivate the name of the unit *group*, we have the following lemma.

Lemma 1.1. Let R be a ring with unity. Then unit group $R^\times$ is a group under multiplication.

Proof. Since $1 \in R$ and $1 \cdot 1 = 1$ we can use 1 as an identity element in $R^\times$. Let u, u' be units in R , then:

$$(1.7) \quad uu'(u')^{-1}u^{-1} = u(u'(u')^{-1})u^{-1} = u \cdot 1 \cdot u^{-1} = 1,$$

$$(1.8) \quad (u')^{-1}u^{-1}uu' = (u')^{-1}(u^{-1}u)u' = (u')^{-1} \cdot 1 \cdot u' = 1.$$

That is, if u and u' are units then so is uu' and the multiplicative inverse of uu' is given by $(u')^{-1}u^{-1}$. In particular, $R^\times$ is closed under $\cdot$ so $(R^\times, \cdot)$ is a set with a binary operation. We already have an identity in $R^\times$ (given by 1), and by the axioms of a ring we know that multiplication is associative, so it remains to show that $R^\times$ have inverse elements. However, by definition of a unit, if $u \in R^\times$ then we have $u^{-1} \in R^\times$ such that $uu^{-1} = u^{-1}u = 1$ (note that if u is a unit then so is u^{-1}), which proves that $(R^\times, \cdot)$ have inverse elements. ■

Lemma 1.2. *Let R be a finite ring with unity and $D \subset R$ the subset of elements that are not zero divisors and not 0. Then $D = R^\times$.*

Proof. Note that if $a, b \in D$ then $ab \in D$. Indeed, assume that $ab \notin D$. Then there is some $c \neq 0$ such that $abc = 0$. If $abc = a(bc) = 0$ then $bc = 0$ since a is not a zero divisor. If $bc = 0$ then $c = 0$ since b is not a zero divisor. However, we assumed that $c \neq 0$ so this is a contradiction. We conclude that $ab \in D$. Let $a \in D$ be arbitrary. We define $f_a : D \rightarrow D$ by $f_a(b) = ab$. If $f_a(b) = f_a(b')$ then $a(b - b') = 0$ which, since $a \neq 0$ is not a zero divisor, implies that $b = b'$. That is, f_a is injective. Since D is finite (it's a subset of the finite set R) we conclude that f_a is also bijective. Now, $1 \in D$ so there is some $b \in D$ such that $ab = f_a(b) = 1$. It remains to show that $ba = 1$ as well (note that we did not assume that R was a commutative ring, if we added this assumption we would be done at this point). To this end, let $ba = c$, after multiplying on both sides with a on the left and recalling that $ab = 1$ we have: $a = ac$, or $a(1 - c) = 0$. Since a is not a zero divisor we conclude that $1 - c = 0$ or $c = 1$ so $ba = c = 1$. That is, we have $ab = ba = 1$ so b is a multiplicative inverse of a and therefore a is a unit. Since a was an arbitrary element of D and $a \in R^\times$ we conclude that $D = R^\times$ (we should also show that $R^\times \subset D$, why does this hold?). ■

Theorem 1.2. *The unit group of $\mathbb{Z}_n^\times$ consist of the elements $j \in \{1, \dots, n-1\}$ that are coprime with n .*

Proof. Recall that the zero divisors in $\mathbb{Z}_n$ are precisely the elements that are not coprime to n , since $\mathbb{Z}_n$ is finite the previous lemma implies the theorem. ■

Definition 1.2. *Let $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ be the function defined by*

$$(1.9) \quad \varphi(n) = \#\{m \leq n \mid \gcd(n, m) = 1\}.$$

This is Euler's totient function (or Euler's phi function).

We can now formulate Euler's generalization of Fermat's little theorem.

Theorem 1.3. *Let $n \geq 1$ be an integer and let $a \in \mathbb{Z}$ be any integer that is relatively prime to n . Then we have $a^{\varphi(n)} = 1 \pmod{n}$.*

Proof. Note that if a is relatively prime to n then $a + n\mathbb{Z} \in \mathbb{Z}_n$ is not a zero divisor. It follows that $a + n\mathbb{Z} \in \mathbb{Z}_n^\times$. Since $\mathbb{Z}_n^\times$ is a group of order $\varphi(n)$ under multiplication, it follows that we have $(a + n\mathbb{Z})^{\varphi(n)} = 1 + n\mathbb{Z}$ or equivalently $a^{\varphi(n)} = 1 \pmod{n}$. ■

1.3. Solving linear equations in $\mathbb{Z}_n$. We will be interested in solving the equation $ax = b$ (with $a, b \in \mathbb{Z}_n$ known and $x \in \mathbb{Z}_n$ the unknown).

Theorem 1.4. *If $a \in \mathbb{Z}_n$ satisfies $\gcd(n, a) = 1$ then $ax = b$ has a unique solution for all $b \in \mathbb{Z}_n$.*

Proof. If $\gcd(n, a) = 1$ then a is not a zero divisor in $\mathbb{Z}_n$ and therefore a unit. We conclude that a has a multiplicative inverse a^{-1} and the equation $ax = b$ can be simplified as $x = a^{-1}b$, which is the unique solution. ■

Theorem 1.5. *If $a \in \mathbb{Z}_n$ satisfies $\gcd(n, a) = d$ then $ax = b$ has a solution if and only if d divides b . In this case the equation has precisely d solutions.*

Proof. Let $d = \gcd(n, a)$. Note that if $ax = b$ in $\mathbb{Z}_n$ then $ax - b$ is divisible by n . Since ax is divisible by d and n is divisible by d it follows that $b = jn - ax$ must also be divisible by d . Conversely, if b is divisible by d then we can write $b = b'd$ and we can write $a = a'd$. So the equation $ax = b$ simplifies as $0 = ax - b = da'x - db' = d(a'x - b')$. Note that $0 = d(a'x - b')$ hold if and only if $a'x - b' = j(n/d)$ for some $j = 0, 1, \dots, d-1$. Since $\gcd(n, a') = 1$ we know that $a'x - b' = j(n/d)$ or equivalently $a'x = b' + j(n/d)$ has a unique solution for each $j = 0, \dots, d-1$. It follows that the original equation $ax = b$ has precisely d solutions corresponding to the solutions to the d different equations $a'x = b' + j(n/d)$. ■

Email address: sandfeldt@uchicago.edu